

Varun Punia

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EDUCATION

Maharaja Agrasen Institute of Technology

Bachelor of Technology - Computer Science and Engineering - GPA : 9/10

Rohini, New Delhi

Dec 2021 – 2025

WORK EXPERIENCE

Machine Learning Intern

Feb 2025 - May 2025

Defence Research and Development Organisation (DRDO)

Metcalfe House, New Delhi

- Worked in the **Scientific Analysis Group (SAG)** lab, conducting **black-box adversarial attacks (FGSM, PGD, Carlini-Wagner)** on **face recognition systems (FaceNet, ArcFace)** to evaluate model vulnerabilities.
- Assessed adversarial robustness by measuring **attack success rates (ASR)**, **perturbation norms (L2/L ∞)**, and **cross-architecture transferability (ResNet, VGGFace2)** via adversarial datasets (LFW, CelebA).
- Boosted model resilience by **23%** using **adversarial training, input preprocessing, and transfer learning (VGGFace2, ResNet)**, preserving baseline accuracy on perturbed inputs.

Software Development Engineer Intern

May 2024 - August 2024

Preplaced

Gurugram, Haryana

- Redesigned the landing page using **Next.js, TypeScript, and Tailwind CSS**, resulting in a notable increase in new user retention from **35.2%** to **51.4%**.
- Revamped the mentor and mentee registration process, **reducing onboarding time by 40%** and **cutting registration errors by 30%**, leading to improved user efficiency and streamlined onboarding.
- Developed and optimized mentor and mentee dashboards, **increasing mentor registrations by 25%** and contributing to a **2% rise** in the likelihood of acquiring **Long-Term Mentee (LTM)**.

Teaching Assistant

Sep 2023 - Mar 2024

Coding Blocks

Pitampura, New Delhi

- Addressed and resolved doubts of **300+** students across **3 batches** on **Data Structures and Algorithms** concepts.
- Compiled a **LeetCode sheet** with **300+** questions covering all topics of **Data Structures and Algorithms**, complete with **expected time and space complexities, editorial and personal solution and hints**.

PROJECTS

StudyNotion - Learn Anything, On Your Schedule [↗](#)

[Live site](#)

- Developed a **scalable ed-tech platform** with a client-server architecture using **React.JS, Node.JS, Express.JS**, and **MongoDB**, implementing **user authentication, course management, Cloudinary media handling, and Razorpay payment integration**.
- Developed **RESTful APIs** for seamless front-end and back-end communication using **JWT** and **Bcrypt**.
- Designed and implemented a **MongoDB NoSQL** database for scalable storage of diverse data types, and optimized the schema using **Mongoose** for **efficient query performance and fast data retrieval**.

Simpli-law - Your AI Legal Assistant [↗](#)

[Live site](#)

- Developed a **legal AI assistant** for **SIH-2023**, integrating **Google authentication, user modes, and secure access features** to provide personalized access to **legal guidance**, including the **Indian constitution, cases & amendments**.
- Enabled support for **10+ Indian regional languages** to ensure accessibility for users from diverse linguistic backgrounds.
- Integrated the **Twingly News API** to provide **real-time legal and judicial news**, keeping users updated.

Ecomzy - Shop With Ease Only on Ecomzy [↗](#)

[Live site](#)

- Created a **responsive e-commerce site** with **React, Redux, and Tailwind CSS** for **optimal user experience**.
- Utilized Redux for efficient **state management** in React, ensuring **scalability and maintainability**.

ACHIEVEMENTS & EXTRACURRICULARS

- Led the team to secure **2nd** position in the **Refine Case Study Challenge**, earning a **cash prize** for good performance.
- Achieved departmental **rank 1** in the 1st semester with a GPA of **9.860**, departmental **rank 10** in the 2nd semester with a GPA of **9.660**, overall **rank 1** in 1st year with GPA of **9.740** and departmental **rank 6** in 6th semester with a GPA of **9.368**
- Achieved **rank 150** in the **International Olympiad of Science**, earning a **silver medal** for exemplary performance.
- Solved **1200+** DSA questions on **LeetCode, GeeksforGeeks, Codeforces** and **Codechef**.
- Selected as **Core Team Lead** in Entrepreneurship & Development Cell (EDC), MAIT thus showing **leadership skills**.
- Cracked **Accenture** and **AccioJob** with a package of **4.5 LPA** and **10 LPA** respectively.

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, JavaScript, TypeScript, SQL, Basics of Java
- **Frontend Development:** HTML, CSS, Bootstrap, Tailwind CSS, DaisyUI, MaterialUI, React.js, Redux, Next.js, Chart.js
- **Backend Development:** Node.js, Express.js, MongoDB Compass, MongoDB Atlas, Mongoose
- **AI & ML:** NumPy, Pandas, Matplotlib, Seaborn, Scikit-Learn, TensorFlow, Keras, PyTorch, FaceNet, ArcFace, OpenCV, Deepface, CNN, BlackBox Attacks (FGSM, PGD, CW), Kaggle, Jupyter Notebook, Google Colab
- **Tools/Technologies:** VS Code, Git, GitHub, Postman, Canva, Notion, Mixpanel, Figma, Cloudinary, Vercel, Render